IChemE

www.icheme.org/cherd
doi: 10.1205/cherd06016

0263–8762/06/$30.00+0.00
© 2006 Institution of Chemical Engineers
Trans IChemE, Part A, October 2006
Chemical Engineering Research and Design, 84(A10): 932–942

ECONOMIC TRADEOFFS INVOLVED IN THE DESIGN OF FERMENTATION PROCESSES WITH ENVIRONMENTAL CONSTRAINTS

G. CORSANO[1]*, O. IRIBARREN[1], J. M. MONTAGNA[1], P. A. AGUIRRE[1] and E. G. SUAREZ[2]

[1]*Instituto de Desarrollo y Diseño INGAR, Santa Fe, Argentina*
[2]*Centro de Análisis de Procesos, Universidad Central Las Villas, Santa Clara, Cuba*

A process model is used for analysing the economic tradeoffs involved in the synthesis, design and operation of a typical batch fermentation plant involving batch and semi-continuous operations with the recycle of the otherwise waste stream that results after the recovery of the product of interest from the fermentation broth. This recycle contributes to a more complete substrate consumption, water reuse and reduction of the environmental impact of the process. Process variables are optimized simultaneously with the plant structure by formulating the whole optimization problem as a non linear programme (NLP). The environmental concern about producing large amounts of fermentation broth waste was accounted for by penalizing its production combined with allowing their recycle to the different types of fermenters based on process considerations. Optimal design and operation are pursued, analysing the economic tradeoffs involved in selecting the number and operation mode of biomass grow and metabolite fermenters, sugar substrate blending to each fermenter, recycle of fermentation broth waste, initial and final concentrations of biomass, substrate and metabolite, and the role of idle times in the process. The paper reports some optimal plant structures and figures for process variables not implemented in industrial practices but supported by process analysis arguments, thus suggesting that they may be worth exploring.

Keywords: mathematical modelling; fermentation; downstream processing; batch processes; optimization; environment.

INTRODUCTION

Batch processes involve batch and semi-continuous operations that need to be scheduled. The constant time and size factors model (Biegler *et al.*, 1997) is used to optimize the plant design by proper selection of the batch sizes, the operating time of semicontinuous units and the structure of the plant (number of units in parallel at each stage and intermediate storage).

If process variables e.g., extents of reactions or reflux ratios are to be optimized too, in a first simpler approach, algebraic process performance models (Salomone and Iribarren, 1992) can be used to describe the time and size factors as functions of the process variables. In this approach the differential equations are analytically integrated, resulting in algebraic equations. The same constant size and time factors model is used, with the process performance models as additional constraints.

**Correspondence to*: Dr G. Corsano, Instituto de Desarrollo y Diseño INGAR, Avellaneda 3657 (S3002GJC), Santa Fe, Argentina.
E-mail: gcorsano@ceride.gov.ar

If more detailed dynamic models are needed to properly describe the batch units, the approach is to discretize the differential equations, as do Bhatia and Biegler (1996) with a reactor–distillation process, or to resort to dynamic simulation modules as do Salomone *et al.* (1996) with fermenters. In both cases the plant structure was fixed.

Barrera and Evans (1989) were the first in proposing an approach for the simultaneous optimization of process variables and the plant structure. They proposed a multilevel approach where the plant structure is at an upper level, the sizing of units at an intermediate level and the optimization of the process variables at a lower level. The approach presented feasibility problems because the optimization at the level of the process variables violated either size constraints imposed at the intermediate level or production targets imposed at the upper level.

Pinto *et al.* (2001) approached the problem for a fermentation process, with a mixed integer non-linear programming (MINLP) model that puts together the MINLP formulation used by other authors to optimize the structure with fixed time and size factor models, adding process

performance models to describe the time and size factors as functions of the process variables.

Asenjo *et al.* (2000) working with the same fermentation process model, proposed to do a first optimization in a one unit per stage–free intermediate storage scenario. Then, they used the values of the process variables at this optimal solution to generate a fixed size and time factors model for optimizing the plant structure, and finally optimized the process variables with this plant structure fixed. Allgor *et al.* (1999) described this multilevel approach as a sequence of *ad hoc* iterations between the structural and process variables optimization sub-problems.

It should be noted that the simultaneous optimization of the structure and process variables for the particular batch reactor–batch distillation process received considerable literature attention, which rendered efficient optimization approaches based on an analysis of the distillation regions.

On the other hand, the fermenters–downstream process did not receive enough attention. There is a considerable number of contributions in the area of synthesis of the sequence of unit operations of the downstream process e.g., in Steffens *et al.* (2000) or in the particular chromatographic protein purification sub-process e.g., in Vasquez *et al.* (2001). But not in deciding the number of fermenter units at each stage and optimizing the size of the fermenters and downstream process subject to scheduling constraints, even if it has a considerable economic impact (Petrides *et al.*, 1995).

In this paper we use the NLP formulation proposed by Corsano *et al.* (2004) for representing the structure of fermentation networks, with the aim of simultaneously optimizing process variables and plant structure of fermentation processes. The process model includes the fermentation stages followed by a downstream process for recovery of the product of interest, with environmental constraints on the waste fermentation broth (Corsano, 2005).

The multiple optimal solutions nature of the problem is explored, finding that the NLP formulation is robust, i.e., the optimal structures arrived at were not dependent on the initial values of the continuous optimization variables. And an analysis is done of the economic tradeoffs involved in selecting the number and operation mode—in parallel or in series—of biomass grow and metabolite fermenters, the sugar substrate blending to each fermenter, the recycle of fermentation broth waste, the initial and final concentrations of biomass and substrate and the role of idle times in the process. This analysis arises from considering the simultaneous optimization of the synthesis, design and operation of the fermentation and downstream processes.

The paper is organized as follows: The study case process is described, afterwards, the models for each unit and their interconnections are presented. Following, is the main contribution of the paper, presenting an analysis of the tradeoffs involved in the optimizations. Finally, the conclusions of this work are presented.

STUDY CASE PROCESS DESCRIPTION

We consider a typical fermentation plant consisting of four stages: biomass production fermentation, metabolite production fermentation, centrifugation of the biomass and a downstream operation for separating the product of interest from the fermentation broth. The process example used here is the production of ethanol, for which a simplified scheme of the process is shown in Figure 1.

The objective of the first batch stage is the production of biomass. The first biomass fermenter, if there were more than one fermenters in series at this stage, is inoculated with a broth containing biomass prepared in laboratory while the next ones are fed by the outlet stream of the preceding unit. At this stage typically large amounts of air are supplied.

The metabolite production stage is also a batch item, fed with the biomass from the first stage and an appropriate fermentation broth. This stage typically works without air supply. We used the yeast and ethanol fermentations superstructure model presented in Corsano *et al.* (2004) to represent these stages. This superstructure model considers the possibility of duplication in series or in parallel for both the biomass and metabolite fermentation stages. The fermenters are modelled by discretizing their mass balances ordinary differential equations. The extents of reaction at each fermenter are process optimization variables.

The final fermentation broth is discharged through a disk stack centrifuge that operates in a semi-continuous mode. The objective of this stage is to separate the biomass from the liquid fermentation broth that contains the metabolite. We used a fixed size factor model for this semi continuous item because its sizing equation does not depend on any of the selected process optimization variables.

The last stage of the process is a batch distillation. The batch distiller is a combination of two batch items: the distiller feed vessel and the distillate tank, and three semi-continuous items: the evaporator, the condenser and the column itself. The analytical process performance model presented by Zamar *et al.* (1998) for batch distillation was adopted, where the size of these items depends on the value of two process optimization variables: the internal reflux ratio and the number of separation stages.

The four stages are connected through the mass balances on the batches that leave one stage to enter into the next. In the case of the fermenters, each one needs an additional stream of substrate. We considered process integration to procure this raw material: two are external carbohydrate sources which are purge streams from a sugar plant i.e., residues of a sugar production process, which can become raw materials of this process: concentrated molasses are the mother liquors of the last sugar crystallization stage and diluted filter juices are the filter permeate of a precipitated phase in an early purification stage of the sugar cane juice.

A third source of carbohydrate is the residual fermentation broth after recovering of the alcohol, that still contains a sugar concentration which we consider a process optimization variable. In this case the residue of the process, known as 'vinasses', can become a raw material of the same process.

The flow sheet of the plant to be optimized feeds each fermenter by blending streams coming from four feed tanks containing: molasses, filter juices, spent fermentation broth and fresh process water. The amounts taken from each source to each fermenter are process optimization variables.

It should be pointed out that from an industrial perspective, this selection of available raw materials implies a

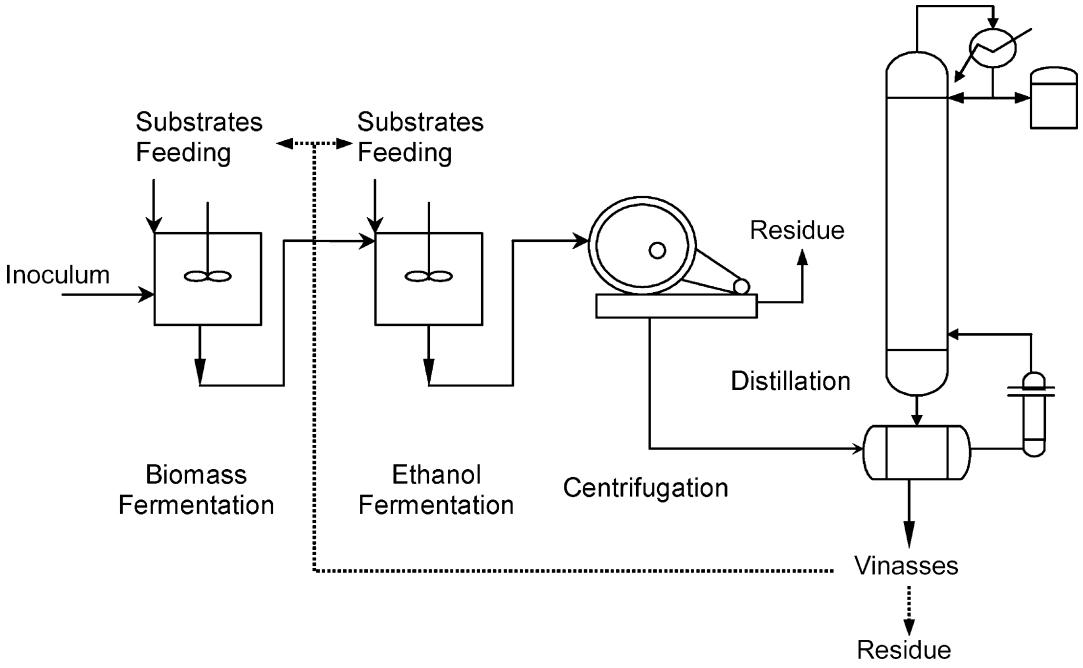


Figure 1. Flowsheet for the ethanol process.

technical risk, i.e., that its implementation in the particular case of ethanol production may face practical drawbacks not considered in this academic analysis. While the use of molasses diluted with fresh process water is the common practice, the use of filter juices is not that common, and using spent fermentation broth is at an experimental stage (De la Cruz Soriano *et al.*, 2003).

We use a combination of technical, economical and environmental factors to handle the issue of favoring or penalizing the use of each sugar source. Technically, we use available industrial data reported by De la Cruz Soriano *et al.* (2003) about the biomass/substrate yield of each source, which is larger for molasses, intermediate for filter juices and lower for spent fermentation broth. Also, given that the ratio of eventually deleterious components to sugar is the largest in the case of spent fermentation broth, we also consider the case of not allowing it as a raw material at the biomass fermenters, but only at the metabolite production fermenters.

Economically, we assign the highest raw material price to molasses, which has a variety of alternative uses and is a concentrated product and thus can be stored, a much smaller raw material price for filter juices, and no cost for the recycled spent fermentation broth.

Environmentally, we penalize not recycled spent fermentation broth with two alternative penalty factors available in the literature (Bates, 1981), proportional to just the amount of residue generated or accounting also for its concentration. While traditionally only molasses and fresh water were used to produce ethanol, the filter juices waste stream from the sugar process and the spent fermentation broth from the ethanol process can also be used as feeds to the biomass and/or ethanol fermentation stages contributing to the sugary substrate of these stages, reducing the consumption of fresh water and reducing the environmental impact of the integrated processes.

The spent fermentation broth substrate concentration and the allocation of the recycles are critical variables for the model, representing a trade off between the size of the fermentation units and the downstream process. Also, as it will be shown, the optimal configuration of biomass and metabolite fermentation stages depends on the spent fermentation broth substrate concentration. These are critical variables in the sense that they produce changes in the optimal process structure.

MATHEMATICAL MODEL OF THE PROCESS

The proposed model consists of a set of equations that describe the mass balances and kinetics at each unit, the units connections constraints, scheduling constraints corresponding to a zero wait transfer policy (ZW) and the objective function equation. The symbols used to describe the process models are defined in the Nomenclature section.

Biomass and Metabolite Fermentations

We adopt the model presented in Corsano *et al.* (2004). For the biomass fermentation stage, equation (4) is removed since in this stage only biomass is produced.

$$\text{Biomass: } \frac{dX_j}{dt} = \mu_j X_j - \upsilon_j X_j \tag{1}$$

$$\text{Substrate: } \frac{dS_j}{dt} = -\frac{\mu_j X_j}{Y_{x/s,j}} \tag{2}$$

$$\text{Dead biomas: } \frac{dX_j^{dead}}{dt} = \upsilon_j X_j \tag{3}$$

$$\text{Product: } \frac{dP_j}{dt} = \frac{\mu_j X_j}{Y_{x/P}} \tag{4}$$

where

$$\mu_j = \mu_{max,\,j}\frac{S_j}{k_{s,j} + S_j} \tag{5}$$

and suffix $j = b, f$ indicates the biomass fermenters set b ($b = 1, \ldots, n_b$) or the metabolite fermenters set f ($f = 1, \ldots, n_f$). Different kinetic coefficients apply depending on the type of fermenter. The maximum specific growth rate of biomass ($\mu_{max,j}$) is $0.5\,h^{-1}$ for biomass fermentation and $0.1\,h^{-1}$ for metabolite production, and for this last fermenter $Y_{x/s,f} = 0.124$ and $Y_{x/P} = 0.23$.

The biomass/substrate yield coefficient $Y_{x/s,b}$ has been experimentally found to depend on the quality of the carbohydrate source, through a nonlinear expression that fits industrial data (De la Cruz Soriano *et al.*, 2003):

$$\begin{aligned} Y_{x/s,b} = {} & 0.3689x_{1b} + 0.4092x_{2b} + 0.2522x_{3b} \\ & + 0.1792x_{1b}x_{2b} + 0.3736x_{1b}x_{3b} + 0.1123x_{2b}x_{3b} \end{aligned} \tag{6}$$

where x_{1b}, x_{2b} and x_{3b} are the sugar contributions of molasses, filter juices and spent fermentation broth respectively, and $x_{1b} + x_{2b} + x_{3b} = 1$.

The differential equations were discretized using the trapezoidal method. The set of discretized differential equations are put together with the algebraic equations that represent the total and components mass balances. Only the first biomass fermenter of the series is inoculated, the remaining units receive the biomass from the previous fermenter. For the first biomass fermenter ($b = 1$):

$$\begin{aligned} x_{1,1}V_1S_{1,initial} = V_{M,1}S_M \qquad x_{2,1}V_1S_{1,initial} = V_{FJ,1}S_{FJ} \\ x_{3,1}V_1S_{1,initial} = V_{W,1}S_W \end{aligned} \tag{7}$$

The equipment interconnection equations are

$$\text{Biomass: } X_{j-1,\,final}V_{j-1} = X_{j,initial}V_j \qquad j = 2, \ldots, n_b,\ 1, \ldots, n_f \tag{8}$$

$$\text{Dead biomass: } X^{dead}_{j-1,final}V_{j-1} = X^{dead}_{j,initial}V_j \qquad j = 2, \ldots, n_b, 1, \ldots, n_f \tag{9}$$

$$\text{Substrate: } \left.\begin{aligned} S_{j-1,\,final}V_{j-1} + S^{feed}_j V^{feed}_j = S_{j,initial}V_j \\ V^{feed}_j = V_{M,j} + V_{FJ,j} + V_{W,j} + V_{Wt,j} \end{aligned}\right\} \qquad j = 2, \ldots, n_b,\ 1, \ldots, n_f \tag{10}$$

$$\text{Product: } P_{j-1,final}V_{j-1} = P_{j,initial}V_j \qquad j = 2, \ldots, n_f \tag{11}$$

$$\text{Volume: } V_{j-1} + V^{feed}_j = V_j \qquad j = 2, \ldots, n_b,\ 1, \ldots, n_f \tag{12}$$

and the initial biomass for the first biomass fermentor is

$$X_{inoc}V_{inoc} = X_{1,initial}V_1 \tag{13}$$

Centrifugation

To separate the biomass, previous to the distillation, a semi-continuous disks stack centrifuge is used. The design equation for this semi-continuous unit is

$$Pw_{cen} = D_{cen}\frac{B_{cen}}{t_{cen}} \tag{14}$$

where Pw_{cen}, D_{cen}, B_{cen} and t_{cen} are the centrifuge size in power units, the centrifuge duty factor, the batch of biomass to be separated and the processing time per cycle. The duty factor was estimated:

$$D_{cen} = \frac{K_C}{X_{fer}\sigma_{cen}}$$

where

$$K_C = \frac{K}{\nu_g} = 0.5\frac{K_W}{m^3/hr} \tag{15}$$

X_{fer} is the final biomass concentration at the metabolite fermentation stage, σ_{cen} is the centrifuge yield measured as the ratio of biomass separated to total biomass present in the feed, K is a constant that depends on the centrifuge type and ν_g is the terminal settling velocity of the particles in a gravitational field. K_C was calculated from data reported by Petrides *et al.* (1995) for *E. coli*, accounting for the relative size between yeast and *E. coli*.

At this stage, the metabolite and substrate concentrations do not change, but the volume and biomass balances are

$$V^{out}_{Wt,cen} = (1 - C_{cen})V^{final}_{Wt,fer} \tag{16}$$

$$V_{Wt,cream} = C_{cen}V^{final}_{Wt,fer} \tag{17}$$

$$V^{out}_{cen} = (1 - C_{cen})V_{fer} \tag{18}$$

$$V_{cream} = C_{cen}V_{fer} \tag{19}$$

$$X^{out}_{cen}V^{out}_{cen} = (1 - \sigma_{cen})X_{fer}V_{fer} \tag{20}$$

$$X_{cream}V_{cream} = \sigma_{cen}X_{fer}V_{fer} \tag{21}$$

The centrifugation dense phase is a cream that can be utilized as cattle feed or recycled to a yeast plant, integrating the processes. C_{cen} is the centrifuge concentration factor.

Distillation

In the distillation of the binary system water–ethanol the minimum boiling temperature azeotrope was taken as the light component, while the water is the heavy component. Let η_l be the light component recovery (amount in the distillate after the separation, divided by the amount in the feed before the separation) and η_h be the heavy component recovery, then

$$d_l = \eta_l f_l \qquad w_l = (1 - \eta_l)f_l \tag{22}$$

$$d_h = \eta_h f_h \qquad w_h = (1 - \eta_h)f_h \tag{23}$$

where $0.8 \le \eta_l \le 0.99$ and $0.001 \le \eta_h \le 0.2$ are taken as the process optimization variables. The model predicts the reflux ratio needed to obtain η_l, η_h in a N stages column. The expression for the minimum number of

stages to perform this separation is

$$N_{\min} = \frac{\ln(\ln(1-\eta_l)/\ln(1-\eta_h))}{\ln(\delta_{l,h})} \tag{24}$$

where $\delta_{l,h}$ is the relative volatility between the azeotrope and water at atmospheric pressure, estimated as $\delta_{l,h} = 2.75$. And the minimum reflux ratio to perform the separation is

$$R_{\min} = \frac{r_l - \delta_{l,h} r_h}{(\delta_{l,h} - 1)} \quad r_l = \frac{\ln(1-\eta_l)}{\ln(1-\eta)} \quad r_h = \frac{\ln(1-\eta_h)}{\ln(1-\eta)} \tag{25}$$

with η being the total recovery $\eta = [(d_l + d_h)/(f_l + f_h)]$.

Finally, the reflux ratio needed to perform the separation $\eta_l \eta_h$ with N stages is

$$R = \frac{R_{\min} + (1 - ((N - N_{\min})/(N+1)))^{2.95}}{1 - (1 - ((N - N_{\min})/(N+1)))^{2.95}} \tag{26}$$

Design equations

The units that compose the distillation facility are sized as follows.

Still vessel: this is a batch item whose size is computed as:

$$V_{SV} = SF_{SV} B \qquad SF_{SV} = \frac{\sum f_i}{(d_h + d_l)\rho_{feed}} \tag{27}$$

where SF_{SV} is the size factor, B the batch size of ethanol produced in kilograms and ρ_{feed} the feed density.

Distillate tank: this is also a batch item that collects the product (ethanol). We compute its size as:

$$V_{DT} = SF_{DT} B \quad SF_{DT} = 1/\rho_{et} \tag{28}$$

where SF_{DT} is the size factor and ρ_{et} is the density of ethanol equal to 790 k m^{-3}.

Evaporator: the evaporation area of this semi continuous item is

$$A_{evap} = D_{evap} \frac{B}{t_{dist}} \qquad D_{evap} = \frac{\Delta H(R+1)}{U_{evap} \Delta T_{evap}} \tag{29}$$

where D_{evap} is the duty factor, $\Delta H = 1046.7$ kJ k^{-1} is heat of vaporization of the azeotrope, $U_{evap} = 4710.15$ kJ (h m^2 °C)$^{-1}$ is the global heat transfer coefficient, $\Delta T_{evap} = 50$°C is temperature difference between water and heating steam, and R is the column reflux ratio. t_{dist} is the processing time for the distillation stage.

Condenser: the condensing area of this semi continuous item is

$$A_{cond} = D_{cond} \frac{B}{t_{dist}} \qquad D_{cond} = \frac{\Delta H(R+1)}{U_{cond} \Delta T_{cond}} \tag{30}$$

where the D_{cond} is the duty factor, $\Delta H = 1046.7$ kJ k^{-1} the heat of vaporization of the azeotrope, $U_{cond} = 2826.1$ kJ (h m^2 °C)$^{-1}$ is the global heat transfer coefficient, $\Delta T_{cond} = 49.3$°C is the logarithmic mean of the temperature differences between condensing azeotrope and cooling water.

Column cross sectional area: we compute the area of this last semi continuous item as

$$A_{trcol} = D_{trcol} \frac{B}{t_{dist}} \qquad D_{trcol} = \frac{R+1}{\rho_G v_G} \tag{31}$$

were D_{trcol} is the duty factor, ρ_G the vapour density $\rho_G = 0.6$ k m^{-3}, v_G is the design vapour velocity in the column, $v_G = 2.5$ m s^{-1} taken from the heuristic $F = v\sqrt{(\rho_G/\rho_L)} = 0.06$ m s^{-1}, where ρ_G and ρ_L are the vapour and liquid densities, respectively. And finally, we assume that the height of a separation stage is 0.6 m.

Mass Balances Connecting Stages

The process model includes a set of equations to account for the mass balances connecting the stages; these include tracking water, biomass, ethanol and substrate transferred from stage to stage when feeding the exit batch from one stage to the downstream one, the options of disposing the spent fermentation broth as a waste stream or recycling it to any of the fermenters, and the optional feed of molasses, filter juices and fresh process water to every fermenter. As a result of the optimization, some of these connections have flux zero, rendering the optimal structure of these connections.

Scheduling Constraints

The transfer policy adopted between batch stages is zero-wait (ZW). That means that the batch processed in a unit is immediately transferred to the following unit. It is worth noting that in this model it is assumed that processing time for batch and semi-continuous units are involved in detailed submodels, some of them written as differential equations and included in the overall model. Duplication of units in series or in parallel out of phase for the biomass and metabolite fermentation stages are alternatives imbedded in the superstructure.

When a batch unit is located between semi-continuous units, the time that the batch unit will be occupied contemplates the material loading time from the previous semi-continuous unit and unloading time for the next semi-continuous unit. For the ethanol process presented in this work, the scheduling constrains are

$$CT_{fer} = \max\left\{\max_{j=b,f}\left[\frac{t_j}{m_j}\right], \frac{t_{nf} + t_{cen}}{m_{nf}}\right\} \tag{32}$$

$$CT_{dist} = t_{cen} + t_{dist} \tag{33}$$

$$CT_{plant} = \max\{CT_{fer}, CT_{dist}\} \tag{34}$$

where m_j is the number of units in parallel out of phase at stage j. It can be noted that the last ethanol fermenter time considers the unload time to the centrifuge (semi-continuous unit) and the distillation time considers the loading time from the centrifuge. Due to the superstructure representation proposed in Corsano *et al.* (2004), the

definition of discrete variables is not needed. Equations (32) and (34) were reformulated in order to avoid a discontinuous NLP by substituting '$\geq$' constraints for 'max' functions.

The production rate of the distilled product Q is a fixed target in this model and this constraint is formulated as

$$Q = \frac{d_{\mathrm{l}} + d_{\mathrm{h}}}{CT_{\mathrm{plant}}} \tag{35}$$

Objective Function

The objective function to be minimized is the total annualized cost (TAC) of the process:

$$MinTAC = CCF \cdot \left(\sum_{j} \alpha_{\mathrm{j}} V_{\mathrm{j}}^{\beta_{\mathrm{j}}} + \sum_{\mathrm{k}} \alpha_{k} V_{\mathrm{k}}^{\beta_{\mathrm{k}}} \right) + OP \tag{36}$$

where j and k are the batch and semi-continuous units respectively, α_{j}, α_{k}, β_{j} and β_{k} are capital cost coefficients according to Petrides *et al*. (1995), V_{j} and V_{k} are the batch and semi-continuous units sizes, CCF is a capital charge factor, $CCF = 0.225$ that considers an amortization term of 0.1 plus 0.125 corresponding to maintenance cost, and OP represents the main operative costs: raw material costs (molasses, filters juices, water), the inoculum cost taken as the cost for producing it at the plant laboratory, the centrifuge operation cost taken as the electricity consumed, and the distillation operation cost that computes the steam and cooling water consumption.

The operative costs also include a penalty for disposal of the spent fermentation broth that depends either on its volume, or on both the volume and its concentration.

Considering volume of the waste stream:

$$Z_{\mathrm{Res}} = Z_{\mathrm{F}} V_{\mathrm{Res}} \frac{HT}{CT_{\mathrm{plant}}} \tag{37}$$

where Z_{Res} represents the total charge for disposal of the residue, V_{Res} is the batch volume of residual spent fermentation broth and Z_{F} is a cost per m^3 of residue. HT represents the production time horizon, which is a problem data.

Considering volume and concentration of the waste:

$$Z_{\mathrm{Res}} = \left(Z_{\mathrm{e}} + Z_{\mathrm{T}} + \frac{S_{\mathrm{w}}}{Os} Z_{\mathrm{BO}} \right) V_{\mathrm{Res}} \frac{HT}{CT_{\mathrm{plant}}} \tag{38}$$

where Z_{e} represents reception and conveyance charge per m^3, Z_{T} volumetric and primary treatment cost per m^3, S_{W} is the residual substrate concentration (k m^{-3}), Os the concentration of settle sewage (k m^{-3}) and Z_{BO} the biological oxidation cost per m^3 of settled sewage. We adopt the average value for $(Z_{\mathrm{e}} + Z_{\mathrm{T}})$ and Z_{BO} from Bates (1981) and $Os = 0.2$ k m^{-3}.

PROCESS OPTIMIZATION, RESULTS AND ANALYSIS

This section presents the results of optimizing the process with the model presented in the previous section. All the examples were implemented and solved in GAMS (Brooke *et al*., 1998) in a Pentium IV, 1.60 Ghz. The code CONOPT2 was employed for solving the NLP problems. The number of variables and constraints of the models solved was about 1200 and the CPU processing times varied between 70 and 90 s.

The study cases presented consider a fixed production rate Q of 100 kg h^{-1} and a time horizon HT of 7500 h, which represents a year. The contribution of total reducing sugars is 779 g l^{-1} for molasses S_{M} and 100 g l^{-1} for filter juices S_{FJ}. The concentration of sugar in the spent fermentation broth S_{W} is a process optimization variable that strongly depends on the last metabolite fermenter operation time. We imposed an upper bound on the initial substrate concentration $S_{\mathrm{j,initial}}$ at every fermenter of 100 g l^{-1} to avoid substrate inhibition, according to industrial practice. The model adopts the following costs for raw materials: inoculums 10 \$ k^{-1}, molasses 31.9 \$ t^{-1}, filter juices 25.9 \$ t^{-1}, fresh water 0.05 \$ t^{-1}.

Optimal Solution

We modelled the problem penalizing the disposal of spent fermentation broth using either equation (37) or (38) but as in both cases the optimal solutions do not discard the spent fermentation broth, the configuration of the plant and the values of the process variables coincide and are presented in Table 1. In Table 2, the processing times for this solution are showed. The optimal plant structure is far from conventional, it has one biomass fermenter and two ethanol fermenters in series. Also far from conventional is the fact that the large optimal value for the substrate concentration of the spent fermentation broth which is 41.4 g l^{-1} and that the second ethanol fermenter has an idle time as shown in Figure 2. Idle time, in this case, refers to the time the fermenter remains empty after finishing a batch and before starting a new operation cycle.

Optimal Solution for Smaller Final Substrate Concentrations

The optimal value for the substrate concentration of the spent fermentation broth reported above is far beyond the figures reported by literature, which range between 10 and 20 g l^{-1}. We next analyse the optimal solution that results from bounding S_{W} to be less than or equal to 20 g l^{-1}.

Again all the produced spent fermentation broth is recycled, so the optimal solution coincides for both equation (37) and (38) penalty cost expressions. The optimal design variables and processing times are displayed in Table 1 and Table 2, respectively. The optimal fermentation configuration consists of two biomass fermenters in series and one ethanol fermenter, the ethanol fermenter has no idle time, and the optimal value for S_{W} is equal to 19.7 g l^{-1} i.e., it is an unconstrained optimal solution that does not reach the upper bound.

So this solution does arrive to a conventional plant structure and also the value for S_{W} in line with the literature reports (De la Cruz Soriano *et al*., 2003). Larger final ethanol concentrations can be obtained if the initial biomass concentration is also high so the best configuration duplicates in series the biomass fermentation stage. In

Table 1. Optimal variables for different optimal solutions.

Optimal S_W	$S_W = 41.4\ g\ l^{-1}$	$S_W = 19.7\ g\ l^{-1}$	$S_W = 10\ g\ l^{-1}$, penalty (37)	$S_W = 10\ g\ l^{-1}$, penalty (38)	$S_W = 36.4\ g\ l^{-1}$, without idle time
Units size					
Biomass fermenter 1 (m^3)	5.77	0.37	0.9	1.1	4.4
Biomass fermenter 2 (m^3)		19.6	15.0	17.3	
Ethanol fermenter 1 (m^3)	18.65	32.9	23.8	23.7	14.5
Ethanol fermenter 2 (m^3)	21.2				23.8
Centrifuge (kwh)	5.27	13.6	9.2	8.5	5.1
Distillation stage number	16	17	17	17	16
Reflux ratio	19.6	31.3	29.5	29.4	22.9
Fermenter blending					
Biomass fermenter 1	Molasses, vinasses, water	Molasses, vinasses	Molasses, vinasses	Molasses, vinasses	Molasses, vinasses
Biomass fermenter 1		Molasses, vinasses	Molasses, water	Molasses, vinasses, water	
Ethanol fermenter 1	Molasses, vinasses, water	Molasses, vinasses, water	Molasses, water	Molasses, vinasses	Molasses, vinasses, water
Ethanol fermenter 2	Molasses				Molasses, vinasses, water
Biomass fermenter 1 yield	0.413	0.402	0.388	0.388	0.41
Biomass fermenter 2 yield		0.402	0.369	0.385	
Produced vinasses ($m^3\ h^{-1}$)	1.23	2.36	1.96	2.05	1.56

Table 2. Processing times for different optimal solutions.

Processing times	$S_W = 41.4\ g\ l^{-1}$	$S_W = 19.7\ g\ l^{-1}$	$S_W = 10\ g\ l^{-1}$, penalty (37)	$S_W = 10\ g\ l^{-1}$, penalty (38)	$S_W = 36.4\ g\ l^{-1}$, without idle time
Biomass fermenter 1	11.8	9.9	8.2	8.2	10.6
Biomass fermenter 2		9.9	8.2	8.2	
Ethanol fermenter 1	11.8	8.5	6.7	5.7	10.6
Ethanol fermenter 2	5.2				7.9
Centrifuge	2.3	1.4	1.5	1.6	2.7
Distillation	9.5	8.5	6.7	6.6	7.9
Plant cycle time	11.8	9.9	8.2	8.2	10.6

this way the cost of inoculum fed to the first biomass fermenter as well as the size of the ethanol fermenter decrease. The fermentation network cost is lower than in the previous case but a more expensive downstream process is required, with a larger total annual cost. Table 3 resumes the operative and investment costs for the different solutions.

Plant Optimal Structure versus Final Substrate Concentration

Figure 3 presents a plot of the total annual cost versus the value of S_W and as can be seen the optimal fermentation structure changes depending on the value of this continuous process optimization variable.

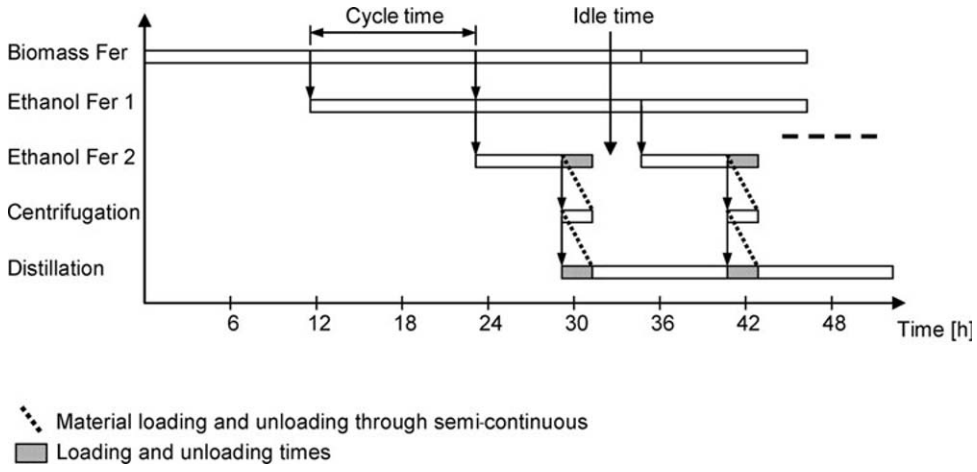


Figure 2. Gantt chart for the optimal solution.

Table 3. Economical comparison between different study cases.

Costs ($ y^{-1})	$S_W = 41.4$ g l^{-1}	$S_W = 19.7$ g l^{-1}	$S_W = 10$ g l^{-1}, penalty (37)	$S_W = 10$ g l^{-1}, penalty (38)	$S_W = 36.4$ g l^{-1}, without idle time
Inoculums	16 950	3075	13 650	16 500	21 375
Raw material	147 150	184 721	216 900	228 150	146 850
Fermentation units	260 700	223 125	224 475	232 275	249 225
Downstream units	145 650	202 500	182 250	182 025	161 250
Vinasses disposal	0	0	7 350	0	0
Energy	51 750	81 150	76 575	76 725	6000
Total annualized	622 200	694 575	721 200	735 675	638 700

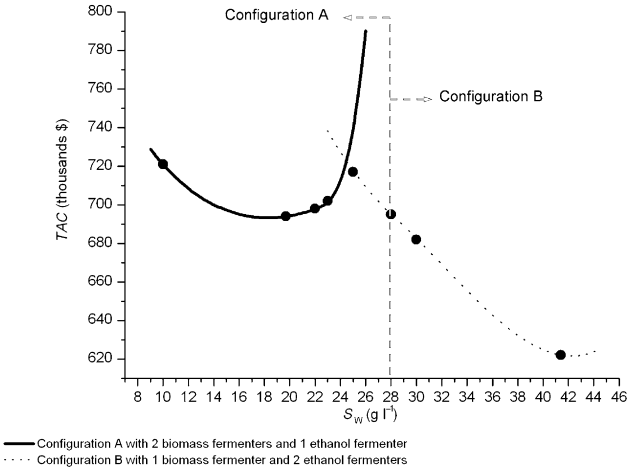


Figure 3. Total annualized cost versus S_W showing the different optimal plant structures.

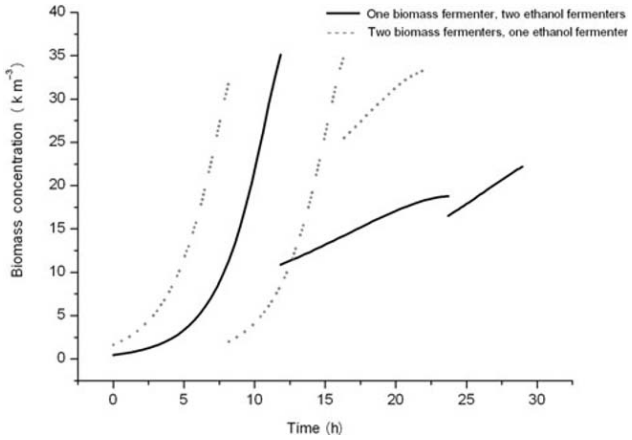


Figure 4. Biomass concentration comparison for different fermentation duplications.

At large values of S_W the optimal solution favours producing a large ethanol concentration feed to the distillation, thus reducing the downstream process cost at the expense of a more expensive fermentation stage with the second ethanol fermenter fed only with molasses, while at low values of S_W the optimal solution finds it more cost effective choosing a structure that optimizes the fermentation stage.

Not only the fermentation network structure changes with S_W but also the optimal recycle structure, so for example at a low final substrate concentration $S_W = 10$ g l^{-1} the optimal fermentation configuration also involves two biomass fermenters and one ethanol fermenter, but the optimal recycle changes depending on the spent fermentation broth disposal cost considered.

When considering equation (37) the optimal solution discards a 96% of the produced spent fermentation broth and the total annualized cost is $721 200. Tables 1 and 2 report the optimal design variables and processing times. For this case the use of water instead of vinasses in the second biomass fermenter improves the yield. This solution has no idle times.

Considering equation (38) instead, all the produced spent fermentation broth is utilized in the fermentation stages. Tables 1 and 2 show the some of the optimal variables for this case. The total annualized cost is $735 675. This cost penalty for disposal forces to use all the produced spent fermentation broth, increasing the size of equipment and the molasses and inoculums consumption.

The reason for using two biomass fermenters in series is to obtain a larger initial biomass concentration at the ethanol fermenter with the result of attaining a larger ethanol concentration. Figure 4 compares biomass concentration in the biomass and ethanol fermenters for the plant structures that duplicate one of these stages. The initial ethanol fermenter biomass concentration is clearly larger when the biomass fermentation unit is duplicated.

The multiple optimal solutions nature of this non convex problem is apparent from the plot in Figure 3 where the optimal solution at $S_W = 19.7$ g l^{-1} is clearly a local one. However, the NLP formulation is robust in arriving to the best solution, i.e., the resulting optimal structure was not dependent on the initial values of the continuous optimisation variables. For example, the optimal solution at $S_W = 41.4$ g l^{-1} reported first, had an initial value $S_W = 12$ g l^{-1}. The global nature of this optimal solution was checked by solving each of the alternative structures embedded in the superstructure, which in this case was possible because its number is relatively small.

Summarizing, when the upper bound for S_W is less than 28 g l^{-1}, the optimal fermentation configuration consists in two biomass fermenters and one ethanol fermenter in series and the optimal value for S_W is equal 19.7 g l^{-1}, regardless of the initial point for this variable. For S_W upper bounded by values greater than 28 g l^{-1} the optimal fermentation configuration consists in one biomass fermenter and two ethanol fermenters in series, and the optimal value of S_W is equal to the upper bound. Finally, when the upper bound for S_W is greater than 42 g l^{-1}, the optimal solution takes the value 41.4 g l^{-1} for this variable. The plotted lines (solid and dotted) on Figure 3 represent the optimal solutions obtaining for fixed values of S_W.

Next, we return to this optimal solution, to analyse the economic tradeoffs involved in the optimization.

Environmental Constraints

Penalizing the disposal of spent fermentation broth results in an optimal input–output structure that recycles this entire stream, with a quite large value for the substrate concentration in the fermentation broth that exits the last fermenter. As the residual stream from the downstream recovery process is not discarded, all its substrate content is recovered. So even if the model contemplates that this sugar has a smaller biomass/substrate yield in the fermenters, the optimal solution favours large end values for the concentration of both substrate and main product, which reduces the cost of the downstream recovery process.

On the other hand, even if the use of filter juice has been favoured in the problem formulation by assigning it a small price compared with molasses, the optimal solution does not consume this raw material. Spent fermentation broth was largely favored to dilute the molasses, followed by fresh process water.

Note that at the optimal solution all the spent fermentation broth is used in the first biomass production fermenter, however using this raw material is known to have a potential deleterious effect on yeast grow not modelled in the present work, and some industries would forbid its use at biomass fermentation stages. That is, do not allow recycle of the spent fermentation broth to the biomass fermentation stage because it inhibits the inoculum.

Adding this prohibition as a constraint to the model we obtain the plant structure with one biomass fermenter and two ethanol fermenters in series either (37) or (38) as penalty functions, but different spent fermentation broth recycle policies. Considering equation (37) the optimal fermentation process variables attain the values displayed in Table 4, and a 17.5% of the total spent fermentation broth produced is discarded. While considering the stronger penalty function (38) the optimal plant structure is the same, the optimal process variable values are displayed on Table 5, and in this case all the produced spent fermentation broth is recycled to the ethanol fermenters, with a total annualized cost greater than in the previous case.

Table 4. Optimal synthesis forbidding vinasses at biomass fermentation, with penalty (37).

	Time (h)	Size
Biomass fermenter	12.6	5.5 (m^3)
Ethanol fermenter 1	12.6	20.7 (m^3)
Ethanol fermenter 2	5.6	23.4 (m^3)
Centrifuge	2.5	5.5 (kwh)
Still vessel		17.4 (m^3)
Distillate tank		1.6 (m^3)
Column cross sectional area	10.1	0.49 (m^2) (D $\cong$ 0.8)
Condenser area		20.0 (m^2)
Evaporator area		11.9 (m^2)
Ethanol final concentration in fermentation stage = 74.5 k m^{-3}		
Number of stages = 16		
Reflux ratio = 20.4		
Substrate vinasses concentration = 43.0 k m^{-3}		
Produced vinasses = 1.27 m^3 h^{-1}		
Residual vinasses = 0.2 m^3 h^{-1}		
Total annualized cost = \$ 638 415		

Table 5. Optimal synthesis forbidding vinasses at biomass fermentation, with penalty (38).

	Time (h)	Size
Biomass fermenter	11.7	4.0 (m^3)
Ethanol fermenter 1	11.7	14.8 (m^3)
Ethanol fermenter 2	9.4	27.3 (m^3)
Centrifuge	2.3	7.0 (kwh)
Still vessel		20.4 (m^3)
Distillate tank		1.5 (m^3)
Column cross sectional area	9.4	0.58 (m^2) (D $\cong$ 0.85)
Condenser area		23.3 (m^2)
Evaporator area		13.8 (m^2)
Ethanol final concentration in fermentation stage = 60.4 k m^{-3}		
Number of stages = 17		
Reflux ratio = 23.9		
Substrate vinasses concentration = 35.4 k m^{-3}		
Produced vinasses = 1.63 m^3 h^{-1}		
Residual vinasses = 0 m^3 h^{-1}		
Total annualized cost = \$ 644 932		

Not allowing the recycle of spent fermentation broth to the biomass fermentation stage makes the optimal solution recycle this stream to the metabolite production fermentation, increasing the cost of the downstream process, mainly due to the final ethanol concentration being smaller.

Idle Time

In the optimal solution reported for the case $S_W = 41.4$ g l^{-1}, the second ethanol production fermenter has an idle time in its cyclic operation, shown in Figure 2. The presence of idle times in batch processes has been traditionally viewed as a non-optimal condition that arises from fixed recipes, so that modelling the stages as a function of process variables would produce optimal solutions with no idle times at all (Douglas, 1988). Later Montagna *et al.* (1994) showed that even though this is usually true when considering only capital costs in the objective function, if operative costs are also considered they may determine that idle times are beneficial. They illustrated the case with a process that consisted of a centrifuge for dewatering crystals followed by a dryer. In this case, an idle time at the centrifuge cycle was convenient because even if a larger operation allowed further dewatering and thus a smaller dryer was required, the economic effect was smaller than the electricity cost of keeping the centrifuge running all the time.

To analyse the economic effect in the present fermentation study case, we replaced the 'greater than' by an 'equal than' scheduling constraint, imposing no idle time. The optimal solution is presented in Tables 1 and 2. Now, the last ethanol fermenter in the series is fed with molasses, water and vinasses, a more diluted blending, and the initial and final concentration values of biomass and product are lower. This requires a more expensive separation task.

Although the total cost of the fermentation stage is lower in this case, the total annualized cost of the overall plant is greater (see Table 3). So, the last fermenter has an idle time because this reduces the downstream costs. The optimal value for S_W is 36.4 g l^{-1} in this case.

Duplication of Units in Series versus in Parallel

Because the kinetics of the ethanol fermentation is slower than the kinetics of the biomass fermentation, without scheduling constraints the ethanol fermenter processing time would be longer than the biomass fermenter processing time. So, due to the scheduling constraints (32) to (34), the duplication of the ethanol fermentation stage could benefit the optimal solution.

Both the duplication in series or in parallel reduce the maximal cycle time of the resulting structure, with their effect being different because duplication in parallel does not change the size factor of the unit while duplication in series does.

As can be observed, all the solutions presented here have units duplication in series at the optimal fermentation synthesis, but not in parallel. And it should be pointed out that duplication in series of biomass fermentations is a common practice, while duplication in series of metabolite production fermentations is not, even if it is well known from chemical engineering fundamentals that a series of batch reactors would in this case demand a smaller total reaction volume than a single batch reactor.

In this study case, even when the reaction kinetics of biomass and ethanol fermentation stages are quite different, a small exponent for fermenters in the capital investment cost formulation penalizes the duplication of units both in series or in parallel.

On the other hand, duplication in series permits the effects on process variables already discussed: duplication in series of biomass fermentation largely increases the amount of biomass fed to the ethanol fermentation, and duplication in series of the ethanol fermentation permits feeding the second fermenter with concentrated molasses, which results in a high ethanol feed to the downstream process.

CONCLUSIONS

We implemented the superstructure optimization for the biomass/ethanol fermentation stages presented by Corsano *et al*. (2004) and completed the fermentation plant description modelling a centrifuge, a batch distillation column and the mass balances model for recycling the spent fermentation broth, to represent a typical fermentation process consisting of a fermentation network and a downstream process to recover the main product.

Design, synthesis and operational issues were simultaneously considered for optimizing the process through a general NLP formulation. Small solution times were required, even when highly non-linear and non-convex formulations were solved.

Integration between units, variable feed blends, unit sizes and operating times, the optimal number of units and the best operating policy—series or parallel—were determined simultaneously with the optimal values of the process variables, and the tradeoffs involved in the optimizations analysed.

The environmental concern about producing large amounts of fermentation broth waste was accounted for by penalizing its production combined with allowing or not their recycle to the different type of fermenters based on process considerations. Depending on the expression utilized for the vinasses disposal cost, the total produced vinasses are partially recycled, totally recycled or totally discarded.

The kinetics and the scheduling constrains for this process model determined that the optimal configuration for biomass and ethanol fermentation stages consists in units duplicated in series.

While duplication in series of biomass fermenters is an industrial practice, duplication in series of metabolite production fermenters and the recycle of spent fermentation broth is not. We are aware that there are deleterious effects of this recycle due to the buildup of other components not considered in this modelling that may jeopardize its ultimate practical impact. We considered only the effect on biomass/substrate yield reported in the literature (De la Cruz Soriano *et al*., 2003) and believe that the modelling of other effects may deserve a specialized research effort as well, that is beyond the scope of this process design analysis. In this sense, the contribution of this work was constructing a model that permitted to envisage that plant structures and process variables figures different from the present industrial practice, may be worth to be explored.

NOMENCLATURE

A	area, m^2
B	batch product, k
C	concentration factor at the centrifuge equal to 0.25
CCF	capital charge factor
CP_{H_2O}	specific heat of cooling water, 4186.8 kJ m^{-3} $^\circ C^{-1}$
CT	cycle time, h
D	duty factor
d	distillate, k
f	distillation feed, k
HT	horizon time, h per year
K_C	centrifugal duty factor constant, 0.5 kw h m^{-3}
$k_{s,j}$	substrate saturation constant, 20 k m^{-3}
m	number of stages in parallel out of phase
N	number of separation stages in the distillation column
OP	operating costs, $
Os	settle sewage concentration, 0.2 k m^{-3}
P	product concentration, k m^{-3}
Pw	centrifuge power, kwh
Q	Production rate, k h^{-1}
r	top–bottom composition ratio
R	reflux ratio
S	substrate concentration, k m^{-3}
SF	Size Factor
t	processing time, h
TAC	total annualized cost
U	global heat transfer coefficient, kJ (h m^2 $^\circ C)^{-1}$
V	volume, m^3
w	distillation waste, k
x_1	percentage contribution of total reducing sugars (TRS) of molasses
x_2	percentage contribution of total reducing sugars (TRS) of filter juice
x_3	percentage contribution of total reducing sugars (TRS) of distillery vinasses
X	biomass concentration, k m^{-3}
$Y_{j,x/s}$	biomass substrate yield coefficient

$Y_{x/P}$	biomass product yield coefficient
Z_{BO}	biological oxidation cost per m^3 of settle sewage, 4.32 \$ m^{-3}
Z_F	cost of m^3 of residual vinasses, 0.5 \$ m^{-3}
Z_e	reception and conveyance charge per m^3
Z_{Res}	total disposal cost
Z_T	volumetric and primary treatment cost per m^3, $Z_e + Z_T = 5.5$ \$ m^{-3}

Greek symbol

α	cost coefficient
β	cost exponent
δ	relative volatility between light and heavy key component
ΔH	azeotrope vaporization heat, 1046.7 kJ k^{-1}
ΔT	temperature difference, evap = 50°C, cond = 49.3°C
ΔT_{cool}	temperature difference for cooling water, 20°C
η	key component recovery, amount in distillate after separation divided by amount in feed before separation
μ	specific growth rate of biomass, h^{-1}
μ_{max}	maximum specific growth rate of biomass, $b = 0.5\ h^{-1}$, $f = 0.1\ h^{-1}$
ρ	density, k m^{-3}
σ	centrifuge constant yield, 0.85
υ	biomass death rate, 0.02 h^{-1}
ν_G	vapour velocity, 2.5 m s^{-1}

Subscripts and superscripts

b	biomass fermenter, $b = b_1, \ldots, n_b$
cen	centrifuge
col	distillation column
cond	condenser
cream	centrifugal residual cream
dead	non-active biomass
dist	distillation process
DT	distillate tank
evap	evaporator
f	ethanol fermenter, $f = f_1, \ldots, n_f$
feed	feed
fer	from fermentation process
final	final component or concentration
FJ	filter Juices
h	heavy key component
initial	initial component or concentration
inoc	inoculum
j	fermentation process units
l	light key component
min	minimum
M	molasses
n_b	number of biomass fermenters
n_f	number of ethanol fermenters
out	outlet stream
plant	plant
Res	residual
SV	still vessel
total	total vinasses
trcol	cross section of distillation column
W	vinasses
Wt	water, m^3

REFERENCES

Allgor, R.J., Evans, L.B. and Barton, P.I., 1999, Screening models for batch process development Part I: design targets for reaction distillation networks, *Chem Eng Sci*., 54: 4145–4164.

Asenjo, J.A., Montagna, J.M., Vecchietti, A.R., Iribarren, O.A. and Pinto, J.M., 2000, Strategies for the simultaneous optimization of the structure and process variables of a protein production plant, *Comput Chem Eng*, 24: 2277–2290.

Barrera, M. and Evans, L.B., 1989, Optimal design and operation of batch processes, *Chem Eng Comm*, 82: 45–66.

Bates, A.J., 1981, *Food Industry Wastes: Disposal and Recovery* (Applied Science Publishers, London, UK).

Bhatia, T. and Biegler, L.T., 1996, Dynamic optimization in the design and scheduling of multiproduct plants, *Ind Eng Chem Res*, 35: 2234–2246.

Biegler, L.T., Grossmann, I.E. and Westerberg, A.W., 1997, *Systematic Methods for Chemical Process Design*. (Prentice-Hall, Englewood Cliffs, NJ, USA).

Brooke, A., Kendrick, D., Meeraus, A. and Raman, R., 1998, *GAMS, A User Guide* (Scientific Press, California, USA).

Corsano, G., Aguirre, P.A., Iribarren, O.A. and Montagna, J.M., 2004, Batch fermentation networks model for optimal synthesis, design and operation, *Ind Eng Chem Res*, 43: 4211–4219.

Corsano, G., 2005, Estrategias en el diseño óptimo de plantas de procesos batch de la industria alimenticia integradas a procesos fermentativos, PhD thesis, Universidad Nacional del Litoral, Argentina.

De la Cruz Soriano, R., González Suárez, E. and Pedraza Gárciga, J., 2003, *Producción de Acetal a Partir de Bioetanol* (Ciencia y Tecnología para el Desarrollo CYTED, Buenos Aires, Argentina).

Douglas, J.M., 1988, *Conceptual Design of Chemical Processes* (MCGraw-Hill, New York, USA).

Montagna, J.M., Iribarren, O.A. and Galiano, F.C., 1994, The design of multiproduct batch plants with process performance models, *Trans IChemE, Part A, Chem Eng Res Des*, 72: 783–791.

Petrides, D., Sapidou, E. and Calandranis, J., 1995, Computer-aided process analysis and economic evaluation for biosynthetic human insulin production—a case study, *Biotechnol Bioeng*, 48: 529–541.

Pinto, J.M., Montagna, J.M., Vecchietti, A.R., Iribarren, O.A. and Asenjo, J.A., 2001, Process performance models in the optimization of multiproduct protein production plants, *Biotechnol Bioeng*, 74(6): 451–465.

Salomone, H.E. and Iribarren, O.A., 1992, Posynomial modeling of batch plants, *Comput Chem Engng*, 16: 173–184.

Salomone, E., Montagna, J.M. and Iribarren, O.A., 1996, Performance analysis of fermentation batch processes by decoupled dynamic simulations, *Process Biochemistry*, 31: 47–61.

Steffens, M.A., Fraga, E.S. and Bogle, I.D.L., 2000, Synthesis of bioprocesses using physical properties data, *Biotechnol Bioeng*, 68: 218–230.

Vasquez, E., Lienqueo, M.E. and Pinto, J.M., 2001, Optimal synthesis of protein purification processes, *Biotechnology Progress*, 17: 685–696.

Zamar, S.D., Salomone, H.E. and Iribarren, O.A., 1998, Shortcut method for multiple task batch distillations, *Ind Eng Chem Res*, 37: 4801–4807.

ACKNOWLEDGEMENTS

The authors want to thank the financial support from CONICET—Consejo Nacional de Investigaciones Científicas y Técnicas—and Agencia Nacional de Promoción Científica y Tecnológica from Argentina.

The manuscript was received 2 February 2006 and accepted for publication after revision 19 May 2006